

PHASE 5: PROJECT DEVELOPMENT PHASE

Project Details

Particular	Details
Project Name	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Phase	Project Development Phase
Date	10 March 2025
Team ID	_____
Prepared By	Koppana Navya

5.1 Data Collection, Loading & Preparation

Introduction

The Project Development Phase began with collecting the Food Ordering Behaviour dataset from a reliable public source. The dataset was imported into Tableau Desktop, where it was cleaned, transformed, and validated before visualization. This ensured the data was accurate, consistent, and suitable for analytical reporting.

Dataset Collection

The Food Ordering Behaviour dataset was downloaded in CSV format from Kaggle. It contains customer ordering information that supports analysis of food ordering trends, customer demographics, restaurant categories, meal preferences, ratings, delivery performance, and revenue.

Dataset Summary

Attribute	Details
Dataset Name	Food Ordering Behaviour Dataset
Source	Kaggle
File Format	CSV
Total Records	50,000

Attribute	Details
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Total Columns	19
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Loading Dataset into Tableau

The dataset was connected to Tableau Desktop using the **Text File** connection option. Tableau automatically recognized the dataset structure, allowing verification of column names and data types.

Steps Performed

- Downloaded the CSV dataset.
 - Connected the file using Tableau Desktop.
 - Verified all fields.
 - Confirmed dimensions and measures.
 - Checked data types.
 - Loaded all 50,000 records successfully.
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Data Preparation

The dataset underwent preprocessing to improve data quality.

Data Cleaning Activities

- Verified column names.
- Removed duplicate records.
- Checked missing values.
- Corrected inconsistent values.
- Standardized categorical data.
- Validated numerical fields.

Calculated Field

A calculated field named **Age Group** was created using the following logic:

Age	Category
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≤20	Child
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Age Category

21–30 Gen Z

31–40 Millennial

>40 Adults

This transformation simplified demographic analysis throughout the project.

5.2 Data Visualization Development

Introduction

After data preparation, Tableau worksheets were developed to visualize customer behaviour, ordering patterns, cuisine preferences, revenue, and delivery performance. Multiple chart types were selected to present the data clearly and effectively.

Developed Visualizations

A total of **nine unique visualizations** were created.

S.No	Visualization	Purpose
1	KPI Cards	Display Total Orders, Revenue, Delivery Fees, and Ratings
2	Heat Map	Analyze cuisine popularity across cities
3	Grouped Bar Chart	Compare age groups and restaurant segments
4	Pie Chart	Show customer preference across cuisines
5	Bar Chart	Analyze company type versus meal type
6	Donut Chart	Compare customer ratings by meal type
7	Revenue Bar Chart	Display revenue contribution by meal type
8	City-wise Bar Chart	Compare order volume across cities
9	Delivery Time Chart	Analyze delivery performance

Dashboard Components

The dashboard includes:

- KPI Cards
 - Interactive Filters
 - City Selection Filter
 - Multiple Charts
 - Dynamic Labels
 - Tooltips
 - Color Legends
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5.3 Dashboard & Story Development

Dashboard Design

After completing individual worksheets, they were integrated into a single interactive dashboard.

Dashboard Features

- Beige/Cream professional theme
 - Title banner
 - Two KPI cards
 - 2×3 chart layout
 - Interactive filtering
 - Responsive design
 - Easy navigation
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Tableau Story

A Tableau Story was created to present the analytical findings in a logical sequence.

Story Scenes

Scene	Description
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Scene 1	KPI Overview
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Scene 2	Cuisine Demand Across Cities
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Scene Description

- Scene 3 Age Group Analysis
- Scene 4 Company Type vs Meal Type
- Scene 5 Customer Rating Analysis
- Scene 6 Revenue Contribution
- Scene 7 Cuisine Distribution
- Scene 8 City-wise Order Volume
- Scene 9 Delivery Time Analysis

The story enables stakeholders to understand customer behaviour through an interactive narrative.

5.4 Development Summary

Technologies Used

Component	Technology
Data Source	CSV Dataset
Data Analytics	Tableau Desktop
Dashboard Publishing	Tableau Public
Web Integration	Flask
Programming Language	Python
Front-End	HTML, CSS

Development Outcomes

The implementation phase achieved the following outcomes:

- Successfully imported and validated 50,000 records.
- Cleaned and transformed the dataset.
- Created calculated fields for customer segmentation.

- Developed nine interactive visualizations.
 - Built a professional Tableau dashboard.
 - Designed a Tableau Story with multiple analytical scenes.
 - Prepared the dashboard for online publishing and web integration.
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Challenges Encountered

- Handling categorical inconsistencies in the dataset.
- Designing visualizations that remained clear with a large number of records.
- Choosing appropriate chart types for different analytical objectives.
- Organizing multiple worksheets into a responsive dashboard layout.

These challenges were addressed through data preprocessing, calculated fields, dashboard optimization, and iterative refinement.

Conclusion

The Project Development Phase transformed the raw Food Ordering Behaviour dataset into an interactive business intelligence solution. Through systematic data preparation, visualization development, dashboard design, and story creation, the project successfully delivered a comprehensive analytical platform capable of uncovering customer ordering trends, cuisine preferences, revenue patterns, and delivery performance. This phase established the core functionality of the project and prepared it for testing, deployment, and end-user access.